Decision Log — Agentic GA4 Analytics

What I assumed and why

- **Correctness and traceability mattered more than unrestricted SQL generation.** GA4's nested events and ambiguous session, purchase, and revenue definitions make plausible answers easy to get wrong. I implemented an 18-metric catalog with server-built, parameterized SQL, retaining guarded ad-hoc SQL for clearly defined questions outside it. Material ambiguity triggers clarification rather than a guessed definition. Findings describe only the available November 2020–January 2021 sample, never lifetime customer behavior.
- **The application must enforce its own limits.** Prompt instructions are guidance, not cost or access controls. The executor checks dataset access, dry-runs queries, enforces execution-time billed-byte caps and a cumulative turn budget, and bounds results. A hand-rolled analyst loop feeds tool errors back for correction; a bounded reviewer checks higher-risk answers before publication.
- **UI libraries were acceptable after clarification.** I asked Mary how strictly to interpret "implement the user interface yourself, without a library," explicitly mentioning React and pre-built component libraries. She confirmed that "React and material ui are both fine." I therefore treated component libraries as permitted and used React with shadcn/Base UI to focus effort on the analytics workflow rather than rebuilding UI primitives.

What I cut or deprioritized

I prioritized a complete question → evidence → chart → narrative → follow-up flow. I deferred specialist-agent delegation, semantic-layer editing, embeddings, additional chart types, and chat deletion. A single analyst plus a limited reviewer provided a smaller, more inspectable design than general multi-agent orchestration. I kept bounded history and exact result retrieval instead of adding potentially lossy conversation summaries. I expanded the scope to include authentication, but deferred administrative tooling and an operational cost dashboard.

Where I got stuck and what I did about it

- **Choosing whether to add a separate planner.** I prototyped a talker–planner–reviewer pipeline to explore whether separating conversation, planning, and review would improve the analysis. In the comparisons I ran, the separate planner did not produce an observable improvement over the simpler analyst–reviewer setup. I therefore deprioritized the three-role pipeline and retained an analyst with a bounded reviewer, avoiding additional orchestration without a demonstrated benefit.

What I would do with another 40 hours

1. **Agent observability and evaluation.** Track model/tool calls, latency, costs, and failures, and expand evaluations to check answer accuracy and appropriate clarification.
2. **Semantic-layer and visualization coverage.** Add high-value metrics with explicit definitions and verified reference results, and expand chart types where they help communicate those metrics.
3. **Dataset extensibility.** Separate GA4-specific schema, metric definitions, and prompt guidance from the agent's orchestration and query execution. Validate those boundaries by integrating a second dataset.
4. **Evidence storage and retrieval efficiency.** Query results are bounded, but their rows are currently duplicated in saved tool responses, chart candidates, and published charts. Store each result once in `query_results`, keep references in messages and chart specifications, and resolve those references into actual rows when building model context or browser responses. Bound conversation reads at the database and fetch only metadata for recent-result discovery, rather than loading full history and results before trimming them.